

Measures of central tendency -

(1)

The value of variable tend to concentrate around the central value. It is the central part of the distribution and therefore they are called the measure of central tendency.

The most common measure of central tendency are:

- 1) Mean or Arithmetic mean
- 2) Median
- 3) Mode
- 4) Weighted Mean
- 5) Combine Mean.

Arithmetic Mean

Mean (or Average) of observations is the sum of the values of all the observations divided by the total number of observations.

Mean for ungrouped data or observation

If $x_1, x_2, x_3, \dots, x_n$ be ' n ' observations for a variable x , the arithmetic mean $\bar{x}$ is given by

$$\text{Mean} = \bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

$$\Rightarrow \boxed{\bar{x} = \frac{\sum x_i}{n} \quad i = 1, 2, \dots, n}$$

Steps of Calculation:-

- ① Add together all the values of x and get $\sum x$
- ② Divide this total by the number of observation.

Example :-

Find the average marks obtained in statistics of 10 students of D BCA.

67, 69, 66, 68, 72, 63, 76, 65, 70, 74.

Soln: $n=10$

$$\begin{aligned}\sum x &= x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} \\ &= 67 + 69 + 66 + 68 + 72 + 63 + 76 + 65 + 70 + 74\end{aligned}$$

$$\Rightarrow \sum x = 690.$$

$$\text{then Mean} = \bar{x} = \frac{\sum x}{n} = \frac{690}{10} = 69$$

$$\bar{x} = 69 \text{ A.}$$

Question The following table contains the half-yearly bonus paid to 10 workers in a factory:

S.No	1	2	3	4	5	6	7	8	9	10
Half-Yearly Bonus	150	200	300	650	250	180	400	500	550	220

② For Discrete series

If $x_1, x_2, x_3, \dots, x_n$ are observations with respective frequencies $f_1, f_2, \dots, f_n$ then

$$\bar{x} = \frac{x_1 f_1 + x_2 f_2 + x_3 f_3 + \dots + x_n f_n}{f_1 + f_2 + \dots + f_n}$$

$$\Rightarrow \boxed{\bar{x} = \frac{\sum x_i f_i}{\sum f_i}}$$

Example

Find the arithmetic mean from the frequency table.

Marks	30	40	50	60	70	80	90
No. of student	15	20	10	15	20	15	5

- ② The marks obtained by 80 students of class X of a certain school in a Mathematics paper consisting of 100 marks are presented in table below. Find the mean of the marks obtained by the student.

Marks obtained x_i	10	20	36	40	50	56	60	70	72	80	88	92	95
No. of student f_i	1	1	3	4	3	2	4	4	1	1	2	3	1

- ③ Calculate the mean of the following frequency distribution of marks in a test in statistics:

Marks	10	20	30	40	50	60	70	80
No. of student	3	6	10	12	9	6	2	2

- ③ Calculation of Arithmetic Mean (Continuous Series).

Let there be a continuous class distribution of the form $x_0 - x_1, x_1 - x_2, x_2 - x_3, \dots, x_{n-1} - x_n$ with the corresponding frequencies $f_1, f_2, \dots, f_n$.

- ① First find mid point of the class interval
② mid point is of each class interval like $m_1, m_2, \dots, m_n$ are the variable x .

class interval	f	mid point	f m
$x_1 - x_2$	f_1	$m_1 = \frac{x_1 + x_2}{2}$	$f_1 m_1$
$x_2 - x_3$	f_2	$m_2 = \frac{x_2 + x_3}{2}$	$f_2 m_2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x_{n-1} - x_n$	f_n	$m_n = \frac{x_{n-1} + x_n}{2}$	$f_n m_n$

then
$$\bar{x} = \frac{f_1 m_1 + f_2 m_2 + \dots + f_n m_n}{f_1 + f_2 + \dots + f_n}$$

$$\Rightarrow \bar{x} = \frac{\sum f_i m_i}{\sum f_i} \quad i = 1, 2, \dots, n$$

Question ①
calculate the arithmetic mean for the following data

class interval	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	3	5	10	15	5	12

② Calculate the arithmetic mean for the following data

class interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Frequency	2	7	17	29	29	10	3	2

③ A class teacher has the following absentee record of 40 students of a class for the whole term. Find the mean number of days a student was absent

No. of days	0-6	6-10	10-14	14-20	20-28	28-38	38-40
No. of students	11	10	7	4	4	3	1

- ④ In a retail market, fruit vendors were selling mangoes kept in packing boxes. These boxes contained varying number of mangoes. The following was the distribution of mangoes according to the number of boxes

No. of Mangoes	50-52	53-55	56-58	59-61	62-64
No. of boxes	15	110	135	115	25

Find the mean number of mangoes kept in a packing box

- ⑤ The table below shows the daily expenditure on food of 25 households in a locality.

Daily expenditure	100-150	150-200	200-250	250-300	300-350
No. of household	4	5	12	2	2